Does syntactic relearning threaten WMDP-unlearned models?

Probably not - for accidental relearning. For deliberate adversarial attacks, the picture is more nuanced.

TL;DR

A recently published attack [5] showed that fine-tuning an unlearned model on structurally similar but topically unrelated text can restore forgotten knowledge 70–100%. I tested whether this applies to WMDP [3], the benchmark used to unlearn bioweapon and cybersecurity knowledge from LLMs. For **accidental relearning through normally distributed fine-tuning data**, it almost certainly does not - WMDP's forget corpus lacks the rigid templates the attack requires. Five text domains including biology-adjacent PubMed literature all fell at or below the similarity level at which the attack is known to fail. However, **deliberate targeted attacks on specific documents remain a demonstrated vulnerability**, and this distinction matters for real-world safety deployment.

Background

Large language models (LLMs) trained on internet-scale data can absorb dangerous knowledge - including information relevant to the synthesis of biological or chemical weapons. One approach to mitigating this risk is **machine unlearning**: modifying a model's weights after training so that it behaves as if it never saw the hazardous content, without retraining from scratch.

The most widely used benchmark for this is **WMDP** (Weapons of Mass Destruction Proxy) [3], which contains 3,668 multiple-choice questions probing hazardous knowledge in biosecurity and cybersecurity. The associated forget corpora - **WMDP-Bio** (PubMed abstracts and full texts) and **WMDP-Cyber** (GitHub documents) - are used to unlearn this knowledge via methods like RMU (Representation Misdirection for Unlearning), GA (Gradient Ascent), and NPO (Negative Preference Optimization).

A natural concern is whether unlearning is durable. If an adversary can fine-tune the unlearned model on benign data and recover the forgotten knowledge, the safety guarantee collapses. This concern was formalised by Hu et al. [2], who demonstrated that relearning on loosely related benign data can reverse unlearning across multiple benchmarks.

The Attack

Yoon et al. [5] demonstrated a striking vulnerability: fine-tuning an unlearned model on data that is **structurally similar but topically unrelated** to the forget corpus can restore supposedly forgotten knowledge 70–100% of the way back to baseline. Crucially, no topical overlap is needed. Sentence structure alone is sufficient.

Their experiments used **TOFU** [4], a synthetic benchmark where 200 fictitious author profiles are memorised and then unlearned. Within the forget corpus, name-format questions follow a rigid, repetitive template - every entry uses the same surface form: *"What is the full name of the author born in [city] on [date]?"* Yoon et al. [5] showed that fine-tuning on retain-set questions with the same template but about different authors caused full knowledge recovery under GA, NPO, and SCRUB unlearning.

Their metric is the **Relearn Success Rate (RSR)**: the fraction of target queries for which the model generates the correct forgotten answer after fine-tuning. Their syntactically similar relearn set achieved Levenshtein similarity of **0.4513** to the target set. The natural question is: does this attack generalise to WMDP [3], where the stakes are much higher?

Why Accidental Relearning Is Hard on WMDP but Easy on TOFU

Yoon et al.'s [5] mechanistic explanation is that gradient ascent-based unlearning primarily suppresses **syntactic template patterns** rather than underlying knowledge tokens. The model learns to avoid generating the surface structure associated with the forget set, but the factual knowledge remains encoded in the weights. Fine-tuning on structurally identical data reactivates those suppressed pathways.

This mechanism requires the forget corpus to have **rigid, repeated templates**. TOFU's [4] name-format questions satisfy this by construction - nine questions all follow the same syntactic skeleton, producing entropy of only **2.95 bits** and top-5 POS skeleton coverage of **66.7%**. Any fine-tuning data using QA-style question templates could inadvertently match this structure.

WMDP [3] is structurally different. Its forget corpus is drawn from real scientific literature - PubMed review articles, experimental papers, and GitHub repositories. Scientific prose does not follow a fixed template. The structural diversity is high and uniform across the corpus: there is **no rigid sub-template to accidentally match**, and no dominant surface pattern that a typical fine-tuning dataset would happen to reproduce.

Methodology

Syntactic diversity was measured using POS (part-of-speech) skeleton analysis implemented with **NLTK** (punkt tokenizer, averaged_perceptron_tagger). Each sentence was reduced to its POS tag sequence, forming a skeleton. For each corpus, 500 sentences were sampled uniformly at random (seed 42) for entropy and top-5 coverage calculations.

For the multi-domain similarity test, Levenshtein similarity between two strings was computed as $1-$(edit distance/max length), following Yoon et al. [5]. Each relearn domain was represented by up to 200 sentences; similarity was measured against a **100-sentence random sample** from the WMDP-Bio forget corpus (seed 42), computing mean pairwise similarity across all domain sentences against all corpus sentences.

All code is available on request. The WMDP-Bio forget corpus requires HuggingFace authentication (`cais/wmdp-bio-forget-corpus`). PubMed domain texts were drawn from the `pubmed_qa` dataset (pqa_unlabeled split). Wikipedia texts from `wikimedia/wikipedia` (20231101.en). Economics and finance texts were generated using the Claude API with explicit register constraints.

Results

Corpus Diversity

Syntactic diversity was measured using Shannon entropy of POS skeleton distributions and top-5 skeleton coverage:

Note: TOFU figures are for the 9 name-format questions constituting the attack's target set (forget05 split). The full forget05 set of 200 questions has entropy 7.62 bits and top-5 coverage 3.5% - similar to WMDP - because the majority of questions are diverse non-name queries. The attack exploits only the rigid name-format sub-template.

WMDP-Bio's entropy of **7.30 bits** and top-5 coverage of **5%** mean the corpus has no dominant syntactic structure - the five most common sentence templates account for only 1 in 20 sentences.

Dataset	Type	Entropy	Top-5 coverage
TOFU name-format [4]	Synthetic QA	2.95 bits	66.7%
WMDP-Bio [3]	Real (PubMed)	7.30 bits	5%
WMDP-Cyber [3]	Real (GitHub)	8.55 bits	10%

Table 1: Syntactic diversity of forget corpora. Higher entropy and lower top-5 coverage indicate greater structural diversity and less exploitable template structure.

Multi-Domain Similarity Test

Yoon et al. [5] provide an empirical reference point in Section 7.1. Their syntactic diversification defence reduces relearn-to-forget similarity from 0.4513 to **0.2241**, at which point RSR stays near zero even after 50 relearning steps. This gives a concrete, empirically validated threshold **below which the attack fails**.

Relearn domain	Similarity to WMDP-Bio	vs safe threshold (0.2241)
Academic economics prose	0.2664	+0.042 above
Wikipedia	0.2314	+0.007 above
Casual personal finance prose	0.2280	+0.004 above
PubMed general biology	**0.2233**	−0.001 **below**
Yoon et al. safe threshold [5]	0.2241	- attack fails here
Yoon et al. attack demonstrated [5]	0.4513	- attack succeeds here

Table 2: Levenshtein similarity of five text domains against the WMDP-Bio forget corpus [3], relative to Yoon et al.'s [5] empirically validated safe threshold of 0.2241.

Four of the five domains fall at or below the safe threshold. **PubMed general biology - the most adversarially plausible domain** - sits at 0.2233, marginally below 0.2241. WMDP's structural diversity provides an implicit version of the syntactic diversification defence without additional engineering.

The Adversarial Threat Model

The findings above address **accidental relearning** - the scenario where a user fine-tunes an unlearned model on normally distributed data and inadvertently recovers dangerous knowledge. For this threat model, the structural diversity of WMDP's [3] forget corpus provides meaningful protection.

> **Known vulnerability: targeted attacks.** WMDP [3] was designed to prevent malicious actors from extracting bioweapon knowledge. A motivated adversary will not rely on accidental relearning. **Yoon et al. [5] demonstrated in Appendix D** that an adversary who identifies specific paragraphs in the forget corpus and applies keyword masking - replacing domain-specific terms with nonsense words while preserving sentence structure - can recover forgotten knowledge with **74% RSR under GA unlearning**. This targeted attack requires access to the forget corpus (publicly available for WMDP) and deliberate document selection, but these are not high barriers for a motivated actor. **This is a known, demonstrated vulnerability that the structural diversity findings here do not address.**

The distinction between accidental and adversarial relearning is important for how WMDP results should be interpreted. **Strong performance under accidental relearning conditions**

does not imply robustness against deliberate attacks. These are separate threat models requiring separate evaluations, a point also raised by Deeb & Roger [1] in the context of broader unlearning limitations.

Implications for Future Benchmarks

The structural difference between TOFU [4] and WMDP [3] reveals a broader methodological problem: the severity of the syntactic relearning vulnerability depends heavily on how the forget corpus was constructed. Synthetic benchmarks with rigid QA templates will always appear more vulnerable than real-world corpora - not because unlearning is more effective on the latter, but because **the attack surface is different**.

This suggests concrete recommendations for future safety benchmarks:

- **Report syntactic diversity metrics** - entropy and top-5 skeleton coverage - as standard metadata alongside any forget corpus, so that vulnerability to template-matching attacks can be assessed without running experiments.
- **Avoid synthetic corpora with rigid templates** for safety-critical benchmarks unless the goal is explicitly to stress-test template suppression. Real scientific corpora with high structural diversity are more representative of deployment conditions.

Limitations

- Five text domains were tested for the similarity analysis. Broader coverage - clinical notes, code, social media - would further strengthen the accidental relearning claim.
- Domain similarity was measured against a **100-sentence random sample** from the 24,453-paper forget corpus. This sample may not fully represent the corpus distribution. Different random samples could produce slightly different similarity scores, and the finding should be interpreted with this sampling uncertainty in mind.
- The entropy and top-5 coverage metrics characterise syntactic diversity **at the corpus level**. They do not measure whether exploitable rigid sub-templates exist within specific documents. As Yoon et al. [5] demonstrated in Appendix D, an adversary only needs a few locally rigid paragraphs - not a globally rigid corpus - to mount a targeted attack. The global diversity finding does not rule out the existence of locally uniform sections that could be identified and exploited by a motivated adversary.

Appendix: Register Experiment

A controlled fine-tuning experiment was run on `cais/Zephyr_RMU` [3]. Two relearn corpora were constructed: 300 sentences of academic economics prose (matched condition) and 300 sentences of casual personal finance prose (control). Both had Levenshtein similarity ≈0.23 to the WMDP-Bio forget corpus and contained no hazardous content. Eight seeds were run per condition using QLoRA (rank=16, lr=2e-4, 3 epochs).

> **Result.** Academic register produced marginally higher WMDP-Bio accuracy recovery: mean gap +1.65pp (matched: 0.3281 ± 0.006, control: 0.3116 ± 0.006). Paired t-test: $t = 3.997$, $p = 0.005$. All 8 seeds showed positive gap direction. Both conditions produced accuracy near random chance (0.25 baseline).

The effect is statistically significant but practically negligible. Whether the gap reflects a genuine register mechanism or floor-level fine-tuning recovery is unclear. This is reported as a preliminary observation for completeness.

References

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